



Home Assignment (HA) -III (CO1)

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CO2: *Understand the various methods like separation of variables and Fourier transform for evaluating the solutions of partial differential equations of real-world problems.*

Q1. Using method of separation of variables solve following partial differential equations:

i) $2u_x + 3u_y + u = 0, u(0, y) = 2e^{-y}$

ii) $\frac{\partial^2 z}{\partial x^2} + \frac{\partial z}{\partial y} = 0, z(0, y) = z(y, \pi) = 0$

Q2. Determine the solution of Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, 0 < x < \pi, 0 < y < \pi$, which satisfies the

Q 3. Solve the integral $\int_0^\infty f(x) \cos sx \, dx = \begin{cases} 1-s, & 0 \leq s \leq 1 \\ 0, & s > 1 \end{cases}$. Also prove that $\int_0^\infty \frac{\sin^2 t}{t^2} \, dt = \frac{\pi}{2}$